Demo 2 Instructions*

EENG350: Electrical Engineering Integration and Design

Prachi Sharma and Tyrone Vincent

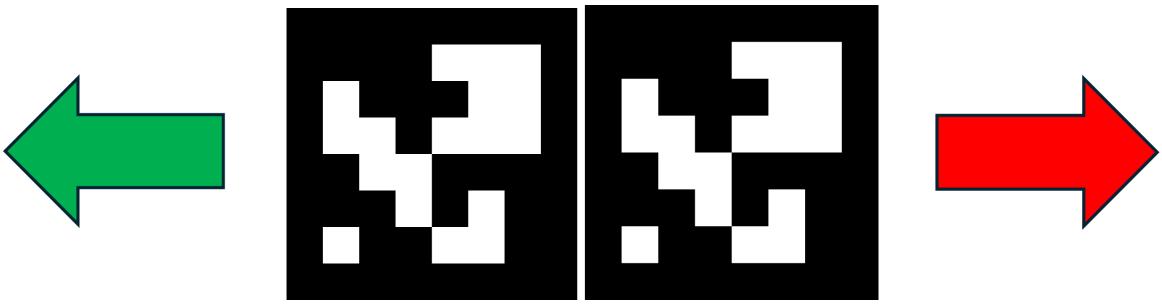
Department of Electrical Engineering
Colorado School of Mines

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1 Demonstration Period 2

Demo 2 is scheduled to be completed by November 13th. Teams that complete their runs on November 13th will receive 100% of their score. Teams that complete on November 16th will receive 90% of their performance score. Teams that complete on November 18th will receive 70% of their performance score. Teams that do not complete by November 18th will receive a zero for their performance score.

In the second demo, you will find a single beacon and turn to the correct direction according to the arrow next to the beacon. The beacons we will be the following:



The marker is in the 6x6 dictionary, and the marker id is 0. The width and length of the marker when printed will be 2 inches.

The beacon will be placed about 5 feet away from the robot (between 4 and 6 feet), while the robot will be given a random initial rotation. There will be two types of trials.

1. In the first type, the robot will rotate to locate the beacon, then move to within 1.5 feet of the beacon and stop before hitting it. The location of the robot is considered to be the center of rotation, or the mid-point between the wheels. If the robot does not start to move within 30 seconds of the go signal, the trial counts as a failure. If the robot moves but fails to stop within 1.5 feet, the trial counts as a failure. The trials are judged based on time to reach the beacon and number of failures before a successful run.

2. In the second type, the robot moves to within 1.5 foot of the beacon, and then turns 90 degrees in the correct direction. If the robot does not start to move within 30 seconds of the go signal, the trial counts as a failure. If the robot hits the beacon, or does not come within 1.5 feet of the beacon before turning, or turns the wrong direction, the trial counts as a failure. The trials are judged based on time and number of failures.

During the test, no communication to the robot is allowed, and the robot must be untethered, so no external connections can be made (e.g. keyboard, usb cable, etc). The teams will go in numerical order. If a team is not ready when they

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are called, they will go to the back of the line. Each team will be given 4 trials, 2 of the first type, and 2 of the second type (one with each beacon). If the team needs more trials in order to get 2 successful trails of each type, they will go to the back of the line. Teams can keep trailing until the end of their class time for their section (10 minutes till the hour).

2 Performance Scoring

The performance metrics for the robot are

- Average time to reach the beacon (based on 2 successful runs)
- Number of failures for first trial
- Average time to reach the beacon and then turn the correct direction (based on 2 successful runs)
- Number of failures for second trial

The score for each category is determined as follows:

- Let B be the target metric, which is the third best score over all sections (e.g: an angle error of 1 degree, and 1 failure)
- Let S be your teams achieved performance metric (e.g. error of 2 degrees, with 2 failures).
- Your category scores are calculated as:
  - Time: $\frac{B}{S} \times 50$ (e.g. $\frac{1}{2} \times 50 = 25$)
  - Failures: $50 - (S - B) \times 10$ (e.g. $50 - (2 - 1) \times 10 = 40$)

The best score is determined on the first day that a team successfully completes a demo. The max in each category is 50 points. Teams earn the sum over all available categories, up to a max of 150 points. The final score is then multiplied by the relevant percentage for the day that the team completes the demo.

3 Documentation

- The reflection logs are to be uploaded to Canvas individually.
- The weekly agenda/minutes should be uploaded to the Canvas "Demo 2 Team Documentation" assignment link. Keep recording your minutes weekly, but you only need to upload the minutes after the demo has been completed.

Your team should be organizing your code and other files in a common repository on github. Create a Demo 2 folder with a readme.md file that describes what is contained there, with further folder divisions as necessary to make everything easy to find. You should include

- Well organized record of all models created in Simulink.
- Well organized Arduino sketches with detailed comments.
- Well organized Python code with detailed comments.

(They don't necessarily have to be organized into folders by language - it could be by task if that makes more sense.) Your github should also have a project board that describes what tasks were assigned and who was responsible.